SHANTANU BAL

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Summary

Senior software engineer with 10+ years of experience building distributed, cloud-native backend systems and large-scale data platforms. Skilled in cloud infrastructure, software design, data structures and algorithms, with experience developing ML platforms, multi-tenant services, and event-driven systems. Collaborative engineer focused on reliable infrastructure and useful customer-facing products.

WORK EXPERIENCE

Axon Enterprise, Inc.
Senior Software Engineer **Sep 2025 - May 2026**

- Designed, built, and shipped cloud services that ingested, enriched, and stored sensor data from field devices at scale (>50M events per day).
- Built a multi-tenant distributed indexing layer that enabled fast searches across large volumes of this sensor data while preserving tenant data ownership and access controls.
- Integrated downstream consumers, including near-real-time hotlist hit alerting pipelines that notified operators of relevant matches.

Amazon.com Services LLC
Senior Software Engineer **May 2022 - Dec 2024**

- Led design and development of the cloud data platform for Amazon Robotics using distributed-system principles, which streamlined the lifecycle management of machine-learning models for over 10K robotic work cells across Amazon's fulfillment network.
- Led development of software supporting machine learning model training, dataset generation and curation, continual retraining, deployment, and real-time performance monitoring.
- Designed sampling strategies for dataset labeling emphasizing diversity and labelling quality; addressed data drift and drove high ML model accuracy in production (mAP, F1).
- Led development of a first-of-its-kind A/B testing infrastructure for robotic work cells, enabling scientists and engineers to evaluate ML models and algorithmic performance.

Software Development Engineer **Jan 2016 - Apr 2022**

- Amazon Robotics: Implemented and integrated Python microservices into the orchestration workflow for robotic work cells, supporting instance segmentation, grasping, and outcome detection.
- Prime Air: Integrated the ship-with-drone option into the amazon.com retail website.
- Alexa - Designed and implemented features for a platform allowing third-party developers to proactively introduce personalized content (monetization/skill upsell, smart-home integration, override system for testing, EU launch in Germany).
- Device Services - Built software platform to bootstrap new devices to Wi-Fi and Amazon accounts via Bluetooth; built iOS and Android libraries for Amazon phone apps to set up IoT devices.

Global Analytics India Ltd.
Software Engineer **May 2011 - May 2012**

- Developed Django (Python) web applications for the company's customer-facing financial products, using Git for version control and SQL databases, which improved transaction processing speeds and user experience.

EDUCATION

Master of Science, Computer Science
University of Southern California *Los Angeles, CA*

Postgraduate Diploma, Management
Indian Institute of Management *Ranchi, India*

Bachelor of Engineering, Computer Science & Engineering
Anna University *Chennai, India*

Skills

Skills: Distributed Systems, Microservice Architecture, Event-Driven Architecture, System Design, Relational/NoSQL Databases, Object-Oriented Programming, Data Structures & Algorithms, Cloud Infrastructure, SQL, DevOps
Technologies & Tools: Python, TypeScript, Java, Docker, Kubernetes, Datadog, Git, Kafka, AWS, DynamoDB, S3, Lambda, SQS, SNS, Step Functions, EC2